



Practical – 10

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Practical Assignment -10

Q. 1. Write algorithm for the following:

a. Write a program that compares two dates (in DD/MM/YYYY format) and prints which one is earlier.

Code:

```
from datetime import datetime
```

```
# Input from user
```

```
date1 = input("Enter first date (DD/MM/YYYY): ")
```

```
date2 = input("Enter second date (DD/MM/YYYY): ")
```

```
# Convert string to datetime object
```

```
d1 = datetime.strptime(date1, "%d/%m/%Y")
```

```
d2 = datetime.strptime(date2, "%d/%m/%Y")
```

```
# Compare and print result
```

```
if d1 < d2:
```

```
    print(f"{date1} is earlier than {date2}")
```

```
elif d1 > d2:
```

```
    print(f"{date2} is earlier than {date1}")
```

```
else:
```

```
    print("Both dates are the same.")
```

```
print("Mehek Sawant S034")
```

Output:

```
==== RESTART: C:/Users/DELL/OneDrive/Desktop/p
Enter first date (DD/MM/YYYY): 01/09/2026
Enter second date (DD/MM/YYYY): 21/06/2026
21/06/2026 is earlier than 01/09/2026
Mehek Sawant S034
> |
```

b. Write a program to measure program execution time.

Code:

```
import time
```

```
# Record start time
```

```
start_time = time.time()
```

```
# sum of numbers upto 1000000
```

```
total = 0
```

```
for i in range(1, 1000000):
```

```
    total += i
```

```
print("Sum of numbers upto 10000: ", total)
```

```
# Record end time
```

```
end_time = time.time()
```

```
# Calculate execution time
```

```
execution_time = end_time - start_time
```

```
print("Execution Time:", execution_time, "seconds")
```

Output:

```
===== RESTART: C:/Users/DELL/OneDrive/Desktop/p
Sum of numbers upto 10000:  499999500000
Execution Time: 0.16560983657836914 seconds
Mehek Sawant S034
```

c. Write a program using the calendar module to print the weekday of the first day of a given month and year.

Code:

```
import calendar
```

```
# Input from user
```

```
year = int(input("Enter year (YYYY): "))
```

```
month = int(input("Enter month (1-12): "))
```

```
# Get weekday of the first day of the given month and year (0=Monday, 6=Sunday)
```

```
weekday = calendar.weekday(year, month, 1)
```

```
# List of Weekday names
```

```
days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday",  
"Sunday"]
```

```
# Print Result using formatted string
```

```
print(f"The first day of {month}/{year} is a {days[weekday]}")
```

```
print("Mehek Sawant S034")
```

Output:

```
===== RESTART: C:/Users/DELL/OneDrive/Desktop/p
Enter year (YYYY): 2026
Enter month (1-12): 6
The first day of 6/2026 is a Monday
Mehek Sawant S034
```

Q. 2. Write a python code snippet to calculate the number of days between two dates.

Code:

```
from datetime import datetime

date1 = input("Enter first date (DD/MM/YYYY): ")
date2 = input("Enter second date (DD/MM/YYYY): ")

d1 = datetime.strptime(date1, "%d/%m/%Y")
d2 = datetime.strptime(date2, "%d/%m/%Y")

days = abs((d2 - d1).days)

print("Number of days between the dates:", days)
```

Output:

```
===== RESTART: C:/Users/DELL/OneDrive/Desktop/pr
Enter first date (DD/MM/YYYY): 01/09/2026
Enter second date (DD/MM/YYYY): 21/06/2027
Number of days between the dates: 293
Mehek Sawant S034
```